

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Total Marks:20

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I Shaik Mahaboob Basha / (192365045) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Mahaboob Basha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well- explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

1. Smart Home Automation System – Types of Intelligent Agents

A smart home automation system controls lighting, temperature, and security by using real-time inputs such as motion sensors, temperature sensors, and time of day. Intelligent agents help the system make decisions automatically.

1. Simple Reflex Agent

- Acts only on the current input using predefined rules.
- Does not remember previous events.

Working in Smart Home:

- If motion is detected → Turn on lights.
- If temperature > 30°C → Turn on air conditioner.
- If door opens at night → Trigger alarm.

Advantages:

- Fast response.
- Easy to implement.

Disadvantages:

- Cannot learn user preferences.
- Cannot handle complex situations.

2. Model-Based Reflex Agent

- Maintains an internal model (memory) of the environment.
- Uses current inputs and past information to make decisions.

Working in Smart Home:

- Remembers whether a room is occupied.
- Keeps track of indoor temperature.
- Knows whether doors and windows are already locked.

Advantages:

- Better decision-making.
- Handles partially observable environments.

Disadvantages:

- More complex than simple reflex agents.

3. Goal-Based Agent

- Makes decisions to achieve specific goals.
- Plans actions before acting.

Working in Smart Home:

- Goal: Maintain comfortable room temperature.
- Goal: Secure the house when everyone leaves.
- Goal: Reduce unnecessary electricity usage.

Advantages:

- Intelligent planning.
- Flexible in different situations.

Disadvantages:

- Requires more computation.
- Slower than reflex agents.

4. Utility-Based Agent

- Chooses the action that provides the highest overall benefit (utility).
- Balances multiple objectives such as comfort, cost, and security.

Working in Smart Home:

- Adjusts AC to maintain comfort while minimizing electricity bills.
- Turns off unused lights to save energy.
- Gives higher priority to security during nighttime.
- Learns user preferences over time for better decisions.

Advantages:

- Optimizes comfort, energy efficiency, and safety.
- Handles conflicting objectives effectively.

Disadvantages:

- More complex to design.
- Requires utility calculations and learning.

Comparison of Intelligent Agents

Agent Type	Memory	Goal-Oriented	Learning	Example in Smart Home
Simple Reflex	No	No	No	Turn on lights when motion is detected
Model-Based	Yes	No	Limited	Remember room occupancy before switching devices
Goal-Based	Yes	Yes	Limited	Lock all doors when leaving home
Utility-Based	Yes	Yes	Yes	Optimize comfort, energy savings, and security together

Best Approach (Justification)

A hybrid approach combining Model-Based, Goal-Based, and Utility-Based agents is the most effective for a smart home automation system.

- Model-Based Agent: Maintains information about occupancy, temperature, and device status.
- Goal-Based Agent: Achieves goals such as maintaining comfort, saving energy, and ensuring home security.
- Utility-Based Agent: Selects the best action by balancing user comfort, energy efficiency, and security while learning user preferences over time.
- Security: Continuously monitors sensors and responds intelligently to potential threats.

Therefore, a hybrid Model-Based + Goal-Based + Utility-Based agent is the most suitable choice for a smart home automation system.

2. Robot in an Unknown Maze – Uninformed Search Strategies

A robot placed in an unknown maze must find a path to the exit without prior knowledge of the maze layout. Uninformed search algorithms such as Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) help the robot explore the maze systematically.

1. Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the lowest cumulative path cost first. It guarantees the least-cost path when movement costs are different.

Working in the Maze:

- The robot calculates the total cost from the start to each position.
- It always explores the path with the lowest cost.
- The search continues until the exit is found.

Advantages:

- Finds the optimal (lowest-cost) path.
- Complete if step costs are positive.
- Suitable for mazes with different movement costs.

Disadvantages:

- High memory usage.
- Slower for large mazes.

Complexity:

- Time: $(O(b^{1+\lfloor C^*/\epsilon \rfloor}))$
- Space: $(O(b^{1+\lfloor C^*/\epsilon \rfloor}))$

Where:

- b = branching factor
- C^* = optimal path cost
- ϵ = minimum step cost

2. Iterative Deepening Search (IDS)

Iterative Deepening Search repeatedly performs Depth-Limited Search, increasing the depth limit until the goal is found.

Working in the Maze:

- Search depth = 0, then 1, then 2, and so on.
- The robot gradually explores deeper levels until it reaches the exit.

Advantages:

- Requires very little memory.
- Complete for finite search spaces.
- Suitable when the depth of the exit is unknown.

Disadvantages:

- Repeats exploration of upper levels.
- Not optimal when path costs vary.

Complexity:

- Time: $(O(b^d))$
- Space: $(O(bd))$

Where:

- b = branching factor
- d = depth of the solution

Comparison of UCS and IDS

Feature	Uniform Cost Search (UCS)	Iterative Deepening Search (IDS)
Strategy	Lowest path cost first	Increase depth limit gradually
Complete	Yes	Yes
Optimal	Yes (positive costs)	Yes (equal step costs only)
Time Complexity	$(O(b^{1+\lfloor C^*/\epsilon \rfloor}))$	$(O(b^d))$
Space Complexity	High	Low
Best Use	Different movement costs	Unknown solution depth

Intelligent Agents in the Maze

1. Simple Reflex Agent

- Acts only on current sensor input.
- May repeatedly visit the same locations.
- Poor performance in an unknown maze.

2. Model-Based Agent

- Maintains a map (memory) of visited cells.
- Avoids revisiting explored paths.
- Makes better navigation decisions.

3. Goal-Based Agent

- Plans actions to reach the exit.
- Uses search algorithms like UCS or IDS to achieve the goal.

4. Utility-Based Agent

- Chooses actions that maximize overall benefit.
- Considers factors such as shortest path, energy consumption, and safety.

Effect of Agent Design on Decision-Making

- Simple Reflex Agent: Fast but cannot remember previous states, leading to inefficient exploration.
- Model-Based Agent: Uses memory to build knowledge of the maze and avoid repeated exploration.
- Goal-Based Agent: Selects actions that move the robot toward the exit using search algorithms.
- Utility-Based Agent: Balances multiple objectives, such as minimizing travel cost, time, and energy.

Best Combination

The Model-Based Goal-Based Agent with Uniform Cost Search (UCS) is the most effective combination.

Justification:

- The Model-Based Agent stores information about visited locations and obstacles.
- The Goal-Based Agent focuses on reaching the exit.
- Uniform Cost Search guarantees the least-cost path when movement costs differ.
- This combination enables efficient navigation, avoids repeated exploration, and finds the optimal route.

If all movements have the same cost and memory is limited, a Goal-Based Agent with Iterative Deepening Search (IDS) is a good alternative because IDS uses much less memory while still guaranteeing a solution.